Lab 6

Aim: To draw Use Case Diagram and write use case description for the project.

1. Users and Use Case Identification

The AssignHub system involves three primary actors, each having clearly defined responsibilities and interactions with the system.

1.1 Student Use Cases

The **Student** is the primary consumer of the assignment submission system and focuses on submitting work and tracking progress.

User Authentication & Profile

- Log in to the system using valid credentials.
- Manage personal profile details.
- View the personalized dashboard.
- Receive system notifications.

Assignment Interaction

- View assignments published for enrolled courses.
- Download assignment reference materials.
- Submit assignment files before the deadline.
- Revise an existing submission if the deadline has not passed.

Tracking & Evaluation

- Track assignment submission status (Not Submitted, Submitted, Late, Evaluated).
- View marks, faculty feedback, and download corrected files.

1.2 Faculty / Instructor Use Cases

The **Faculty** actor manages the complete lifecycle of assignments and student evaluations.

Assignment Management

- Create new assignments with descriptions and deadlines.
- Publish or unpublish assignments to control student visibility.
- View student submissions for assigned courses.

Evaluation & Feedback

- Evaluate submitted assignments.
- Assign marks to student submissions.
- Upload corrected assignment files.

Administrative Control

- Extend assignment deadlines for specific students or batches.
- Generate reports and view evaluation statistics through the dashboard.

1.3 Administrator Use Cases

The **Administrator** oversees system configuration, user management, and academic structure.

User Account Administration

- Create new user accounts.
- Update existing user details.
- Deactivate user accounts when required.

Academic Structure Management

- Create and manage courses.
- Create and manage batches.
- Assign faculty to courses and batches.

System Oversight

- Configure system-level settings.
- View audit logs for monitoring and security purposes.

2. Detailed Use Case Descriptions - AssignHub

1. Use Case Name: Authenticate User (Login)

Actor: User (Student / Faculty / Administrator)

Description: Allows a registered user to securely log into the system and access their role-based dashboard.

Precondition:
- User account exists in the system.
- User is not already authenticated.

Postcondition:
- User session is created.
- User is redirected to the appropriate dashboard.

Relationships:
- <<include>> Validate Credentials
- <<include>> Identify User Role
- Generalization: Student, Faculty, Administrator inherit from User

Flow:

Basic Flow:
1. User enters username/email and password.
2. System validates credentials.
3. System identifies user role.
4. System redirects to role-specific dashboard.

Alternate Flow:
- If credentials are incorrect → system displays error message.

Exception Flow:
- If account is deactivated → access denied.

Notes: Authentication is mandatory before accessing any other system functionality.

2. Use Case Name: Submit Assignment

Actor: Student

Description: Allows the student to upload and submit an assignment file before the deadline.

Precondition:
- Assignment is published.
- Deadline has not expired.
- Student is authenticated.

Postcondition:
- Submission is stored with timestamp.
- Submission status updated to “Submitted”.

Relationships:
- <<include>> Validate File
- <<include>> Store Submission

Flow:

Basic Flow:
1. Student selects assignment.
2. Student uploads assignment file.
3. System validates file format and size.
4. System stores submission.
5. Confirmation message displayed.

Alternate Flow:
- If file format invalid → error message shown.

Exception Flow:
- If deadline passed → submission marked as “Late” or rejected based on policy.

Notes: Submission timestamp is automatically recorded.

3. Use Case Name: Revise Submission

Actor: Student

Description: Allows the student to replace an existing submission before the deadline.

Precondition:
- Submission already exists.
- Deadline not expired.

Postcondition:
- Old submission replaced with new file.
- Updated timestamp recorded.

Relationships:
- <<extend>> Submit Assignment

Flow:

Basic Flow:
1. Student selects submitted assignment.
2. Student uploads new file.
3. System replaces previous file.
4. Confirmation displayed.

Exception Flow:
- If deadline expired → revision not allowed.

Notes: This is an optional extension of Submit Assignment.

4. Use Case Name: Create Assignment

Actor: Faculty

Description: Allows faculty to create a new assignment for a specific course and batch.

Precondition:
- Faculty is authenticated.
- Faculty is mapped to course and batch.

Postcondition:
- Assignment stored in system.

Flow:

Basic Flow:
1. Faculty enters title, description, dates, and marks.
2. System validates inputs.
3. Assignment is saved.
4. Confirmation displayed.

Alternate Flow:
- If required fields missing → validation error.

Exception Flow:
- If database error occurs → assignment not saved.

Notes: Assignment remains unpublished until explicitly published.

5. Use Case Name: Evaluate Assignment

Actor: Faculty

Description: Allows faculty to review student submissions, assign marks, and provide feedback.

Precondition:
- Student submission exists.
- Faculty is authenticated.

Postcondition:
- Marks and feedback stored.
- Submission status updated to “Evaluated”.

Relationships:
- <<include>> Assign Marks
- <<include>> Upload Corrected File

Flow:

Basic Flow:
1. Faculty selects student submission.
2. Faculty assigns marks.
3. Faculty enters remarks.
4. Faculty uploads corrected file (optional).
5. System saves evaluation.

Alternate Flow:
- Faculty may skip uploading corrected file.

Exception Flow:
- If submission not found → error displayed.

Notes:
Evaluation triggers notification to student.

6. Use Case Name: Manage Users

Actor: Administrator

Description: Allows administrator to create, update, and deactivate user accounts.

Precondition:
- Administrator is authenticated.

Postcondition:
- User account data updated in system.

Relationships:
- Generalization: Create User, Update User, Deactivate User

Flow:

Basic Flow:
1. Admin selects user management option.
2. Admin enters or modifies user details.
3. System validates input.
4. Changes saved.

Alternate Flow:
- If duplicate user detected → error shown.

Exception Flow:
- If system error occurs → changes not saved.

Notes: User data is not permanently deleted; only deactivated.

7. Use Case Name: Manage Academic Structure

Actor: Administrator

Description: Allows administrator to manage courses, batches, and faculty-course-batch mappings.

Precondition:
- Administrator is authenticated.

Postcondition:
- Academic data updated.

Relationships:
- Generalization: Manage Courses, Manage Batches, Assign Faculty to Course & Batch

Flow:

Basic Flow:
1. Admin creates or updates course.
2. Admin creates batch.
3. Admin assigns faculty to course & batch.
4. System stores mappings.

Alternate Flow:
- If course already exists → update instead of create.

Exception Flow:
- If invalid mapping → system rejects request.

Notes: This use case ensures proper academic organization.

8. Use Case Name: Extend Deadline

Actor: Faculty

Description: Allows faculty to extend assignment deadline for specific students or batches.

Precondition:
- Assignment exists.
- Faculty is authenticated.

Postcondition:
- New deadline applied.

Relationships:
- <<extend>> Publish / Unpublish Assignment

Flow:

Basic Flow:
1. Faculty selects assignment.
2. Faculty selects student or batch.
3. Faculty sets new deadline.
4. System updates deadline.

Exception Flow:
- If assignment not found → error displayed.

Notes: Deadline extension overrides default deadline for selected users only.

9. Use Case Name: View Assignments

Actor: Student

Description: Allows the student to access & view all assignments published for their enrolled courses, including assignment details and deadline.

Precondition:
- Student is authenticated.
- Student is enrolled in at least one course.
- Assignments are published by faculty.

Postcondition:
- Assignment list is displayed.
- Student can view assignment description and related resources.

Relationships:
- Generalization: Student inherits User.

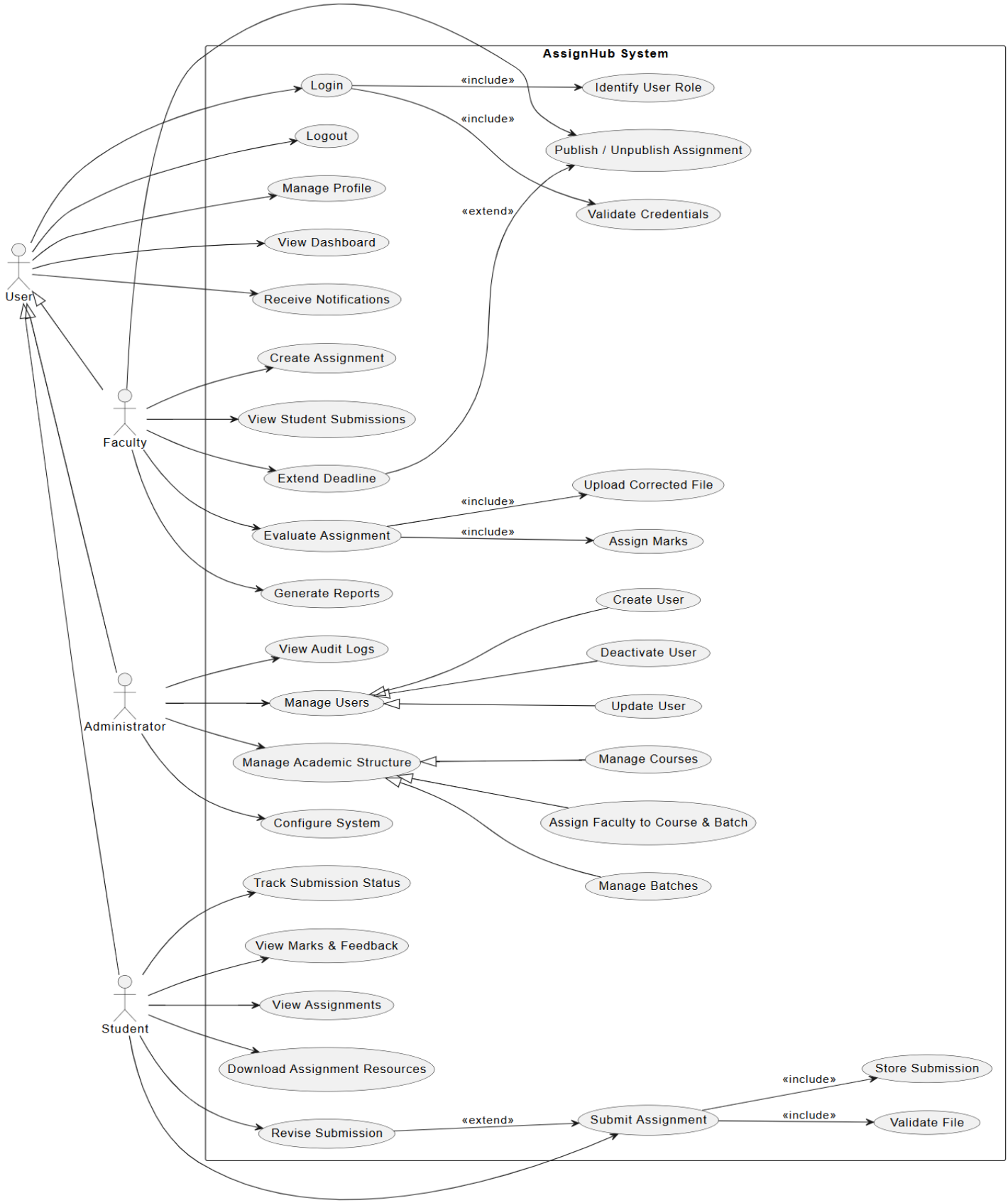
Flow:

Basic Flow:
1. Student logs into the system.
2. Student selects "View Assignments" option.
3. System retrieves assignments associated with the student's enrolled courses.
4. System displays assignment title, description, start date, due date, and submission status.

Alternate Flow:
- If no assignments are available → system displays "No assignments found."

Exception Flow:
- If database retrieval fails → system displays error message.

Notes: This use case is essential before performing Submit Assignment. It ensures students only access assignments relevant to their enrolled courses.


AssignHub System
User
Faculty
Administrator
Student
Login
Logout
Manage Profile
View Dashboard
Receive Notifications
Identify User Role
Publish / Unpublish Assignment
Validate Credentials
«include»
«include»
«extend»
Create Assignment
View Student Submissions
Extend Deadline
Evaluate Assignment
Upload Corrected File
Assign Marks
«include»
«include»
Generate Reports
View Audit Logs
Manage Users
Create User
Deactivate User
Update User
Manage Academic Structure
Manage Courses
Assign Faculty to Course & Batch
Manage Batches
Configure System
Track Submission Status
View Marks & Feedback
View Assignments
Download Assignment Resources
Revise Submission
«extend»
Submit Assignment
Store Submission
Validate File
«include»
«include»